Formulas and Variable Definitions

Renewable Energy Sources: PYQ Companion

A concise reference to the formulas used in the 21 solved numericals. Units and symbol meanings are local to each topic; all trigonometric angles are evaluated in degrees.

1. Solar geometry and solar time

Declination, sunset hour angle and daylight duration

$$\delta \simeq 23.45^\circ \sin\left[\frac{360^\circ(284+n)}{365}\right], \qquad \omega_s = \cos^{-1}(-\tan\phi\tan\delta),$$
$$N = \frac{2\omega_s}{15} \quad \text{hours, with } \omega_s \text{ in degrees.} \tag{1}$$

Hour angle, collector incidence and solar zenith

$$\begin{aligned} \omega &= 15^\circ(t_s - 12), \\ \cos\theta &= \sin\delta\sin(\phi-\beta) + \cos\delta\cos(\phi-\beta)\cos\omega, \\ \cos\theta_z &= \sin\phi\sin\delta + \cos\phi\cos\delta\cos\omega. \end{aligned} \tag{2}$$

The collector-incidence expression applies to a surface facing due south; t_s is expressed in hours.

Clock-time correction and equation of time

$$B = \frac{360^\circ(n-81)}{364}, \qquad E = 9.87\sin 2B - 7.53\cos B - 1.5\sin B,$$
$$t_s = t_{\text{std}} + \frac{4(\lambda - \lambda_{\text{st}}) + E}{60}. \tag{3}$$

Here E is in minutes, longitudes are positive east, and both times are in hours. The coefficient 4 converts one degree of longitude into four minutes of time.

Symbol	Variable name, meaning and units
n	Day number of the year: January 1 is $n = 1$; count February 29 in a leap year.
ϕ, δ	Geographic latitude and solar declination, respectively (°); north is positive.
ω, ω_s	Solar hour angle and sunset hour angle (°). Morning $\omega < 0$, afternoon $\omega > 0$; sunset $\omega_s > 0$.
N	Geometric daylight duration (h).
β	Collector tilt measured from the horizontal (°).
θ, θ_z	Beam-incidence angle from the collector normal, and solar zenith angle from the local vertical (°).
t_s, t_{std}	Local apparent solar time and standard clock time (decimal h).
λ, λ_{st}	Local longitude and standard-time meridian longitude (° E).
B, E	Auxiliary annual angle (°) and equation of time (min). Here E = apparent solar time minus local mean solar time.

Conventions. The declination and equation-of-time relations are the same engineering approximations used in the solutions. Geometric day length excludes refraction. Actual IST uses $\lambda_{\text{st}} = 82.5^\circ$ E; a different printed meridian must be identified explicitly.

2. Wind turbines and shaft torque

HAWT denotes a horizontal-axis wind turbine; VAWT denotes a vertical-axis wind turbine.

Wind profile, swept area and air density

$$\boxed{\begin{aligned} V_1 = V_{\text{ref}}\left(\frac{z}{z_{\text{ref}}}\right)^{\alpha}, \qquad A_{\text{HAWT}} = \frac{\pi D^2}{4}, \qquad A_{\text{VAWT}} = DH, \\ \rho = \frac{p}{R_{\text{air}}T}, \qquad T\,[\text{K}] = T_C\,[^\circ\text{C}] + 273.15. \end{aligned}} \tag{4}$$

Velocity reduction, induction and power

$$\boxed{\begin{aligned} k = \frac{V_2}{V_1}, \qquad V_d = \frac{V_1 + V_2}{2}, \qquad a = 1 - \frac{V_d}{V_1} = \frac{1-k}{2}, \\ C_p = 4a(1-a)^2, \qquad P_w = \frac{1}{2}\rho A V_1^3, \qquad P_t = C_p P_w, \qquad P_e = \eta_g P_t. \end{aligned}} \tag{5}$$

Axial force, maximum power and torque

$$\boxed{\begin{aligned} F = \frac{1}{2}\rho A(V_1^2 - V_2^2) = \frac{1}{2}\rho A V_1^2(1-k^2), \\ P_{\max} = \frac{16}{27}P_w, \qquad \Omega = \frac{2\pi N_r}{60}, \qquad \tau = \frac{P_t}{\Omega}, \quad \tau_{\max} = \frac{P_{\max}}{\Omega}. \end{aligned}} \tag{6}$$

Symbol	Variable name, meaning and units
V_1	Upstream, undisturbed wind speed at hub height (m s^{-1}).
V_{ref}, V_d, V_2	Reference-height wind speed, speed at the rotor plane, and far-downstream wind speed, respectively (m s^{-1}).
z, z_{ref}	Hub height and reference measurement height above ground (m).
α	Wind-shear exponent in the power-law velocity profile (dimensionless).
A, D, H	Swept area (m^2), rotor diameter (m), and VAWT blade height (m). Tower height is not blade height.
ρ, p	Air density (kg m^{-3}) and absolute air pressure (Pa).
T, T_C	Absolute air temperature (K) and air temperature in degrees Celsius (°C).
R_{air}	Specific gas constant for dry air: $287\,\text{J kg}^{-1}\,\text{K}^{-1}$.
k, a	Downstream-to-upstream speed ratio and axial induction (interference) factor (dimensionless).
C_p, η_g	Turbine power coefficient and generator efficiency (dimensionless fractions).
P_w, P_t	Power available in the undisturbed wind and mechanical power extracted by the turbine (W).
P_e, $P_{\max}$	Electrical generator output and ideal Betz-limit maximum mechanical power (W).
F	Axial force or thrust on the rotor (N).
N_r, Ω	Rotor rotational speed (rpm) and angular speed (rad s^{-1}).
τ, $\tau_{\max}$	Shaft torque and ideal maximum shaft torque (N m).

Use. These relations assume an ideal actuator disc. A reduction *at the rotor* specifies V_d; a *downstream* ratio specifies V_2. For torque, substitute power in watts. 1 atm = 101 325 Pa. The notation N_r here denotes the rotor speed written as N in the torque solutions, distinguishing it from solar daylight duration.

3. Project finance and biogas sizing

Present value and net present value

$$\boxed{PV_t = \frac{C_t}{(1+i)^t}, \qquad \mathrm{NPV} = -C_0 + \sum_{t=1}^{n} \frac{C_t}{(1+i)^t}.} \tag{7}$$

Symbol	Variable name, meaning and units
C_0, C_t	Initial investment at year zero and net cash inflow at the end of year t (Rs.).
i, t, n	Annual discount rate (fraction), cash-flow year index, and project duration (years).
PV_t, NPV	Present value of the year-t inflow and net present value of the project (Rs.).

Decision. A positive NPV is acceptable at the specified rate. Among comparable alternatives, prefer the higher NPV; two negative NPVs do not justify accepting either project.

Daily gas demand and number of cows

$$\boxed{G = N_p g_c + N_L t_L g_L, \qquad G_{\mathrm{cow}} = m_{w,\mathrm{cow}} f_c f_d Y, \qquad N_{\mathrm{cow}} = \left\lceil \frac{G}{G_{\mathrm{cow}}} \right\rceil.} \tag{8}$$

Daily feed, digester volume and thermal power

$$\boxed{\begin{aligned} m_d &= \frac{G}{Y}, \qquad m_w = \frac{m_d}{f_d}, \qquad V_{\mathrm{dig}} = \frac{(m_w + m_{\mathrm{water}})\,\mathrm{HRT}}{\rho_s f_o}, \\ G_{\mathrm{produced}} &= m_d Y, \qquad P_{\mathrm{gas}} = \frac{G_{\mathrm{produced}}\, HV\, 10^6}{86\,400}, \qquad P_{\mathrm{useful}} = \eta_b P_{\mathrm{gas}}. \end{aligned}} \tag{9}$$

For a specified herd, $m_d = N_{\mathrm{cow}} m_{d,\mathrm{cow}}$ when all stated dry matter is fed. When raw fresh-dung production is given, $m_w = N_{\mathrm{cow}} m_{w,\mathrm{cow}} f_c$ and $m_d = f_d m_w$.

Symbol	Variable name, meaning and units
G, G_{produced}	Required and produced gas volume per day ($\mathrm{m^3\,day^{-1}}$).
N_p, N_L, t_L	Number of adults, number of lamps, and daily operating time per lamp ($\mathrm{h\,day^{-1}}$).
g_c, g_L	Cooking-gas demand per adult ($\mathrm{m^3\,person^{-1}\,day^{-1}}$) and consumption per lamp ($\mathrm{m^3\,h^{-1}}$).
$m_{w,\mathrm{cow}}$, $m_{d,\mathrm{cow}}$	Daily fresh-dung production and daily dry matter per cow, respectively ($\mathrm{kg\,day^{-1}}$ per cow).
f_c, f_d	Collectible fraction of fresh dung and dry-matter mass fraction of fresh dung (dimensionless).
Y, G_{cow}	Gas yield per kg of dry matter ($\mathrm{m^3\,kg^{-1}}$) and usable daily gas per cow ($\mathrm{m^3\,day^{-1}}$ per cow).
N_{cow}	Required whole number of cows; $\lceil x \rceil$ means round x upward to an integer.
m_d, m_w, m_{water}	Daily dry-matter feed, fresh-dung feed, and added dilution water ($\mathrm{kg\,day^{-1}}$).
ρ_s, f_o	Slurry density ($\mathrm{kg\,m^{-3}}$) and fraction of total digester volume occupied by slurry (dimensionless).
HRT, V_{dig}	Hydraulic retention time (days) and total digester volume ($\mathrm{m^3}$).
HV, η_b	Biogas heating value ($\mathrm{MJ\,m^{-3}}$) and burner efficiency (dimensionless fraction).
P_{gas}, P_{useful}	Average gross thermal power and useful burner output (W).

Sizing conventions. For 1:1 fresh dung : added water by mass, $m_{\mathrm{water}} = m_w$; use $f_o = 0.90$ when 90% occupancy is specified. The thermal-power equation uses $1\,\mathrm{MJ} = 10^6\,\mathrm{J}$ and $1\,\mathrm{day} = 86\,400\,\mathrm{s}$. Efficiencies and percentages are substituted as fractions.